



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to R.V. VIKRAM AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

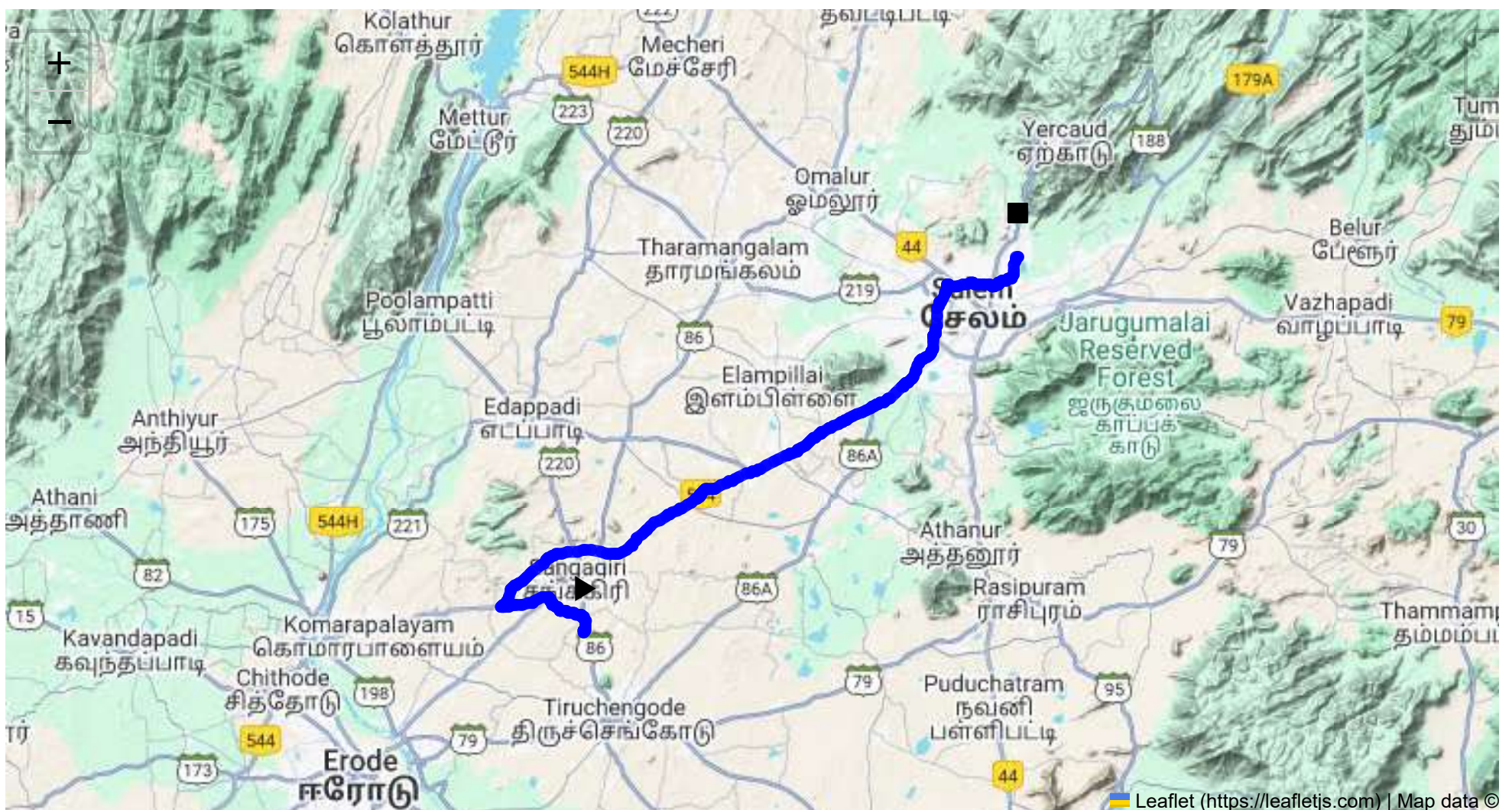
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 60.55 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.695788, 78.176514)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W (Sangagiri) to No45, Kannankurichi Main Rd (Salem) via Sankari RS, Sankari West, and Padaiveedu follows a path through several district roads, primarily on Bhavani Main Road. This approximately 60.55 km journey moves through semi-urban and rural environments with connections to major highways in Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with:

- Summer (March to June): High temperatures up to 40°C. Potential heatwaves may cause road surface degradation.
- Monsoon (July to September): Heavy rainfall can lead to waterlogging and poor visibility; some flooding in low-lying areas.
- Winter (December to February): Generally mild weather, minimal rainfall risks.

3. Analysis of Traffic Patterns

- **Peak Hours:** Considerable traffic congestion during morning (8:00 AM - 10:00 AM) and evening rush hours (5:00 PM - 7:00 PM), particularly near urban sections.
- **Congestion-Prone Areas:** Sankari RS junctions and entry points to Salem can experience significant delays due to local commercial activities and school zones.

4. Assessment of Road Quality and Infrastructure

- Roads are mostly paved but may have uneven surfaces and pothole-prone segments, especially following monsoon seasons.
- Lane markings may sometimes be faded, and lighting is variably present along the entire stretch, affecting night driving.

5. Suggestions for Alternative Routes for Emergencies

- During significant delays or road closures, consider NH544 or NH44 as potential alternate highways, though this will extend the route, especially towards the southern and eastern sections.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Specific permits required for transporting hazardous materials, with restrictions on transporting certain chemicals through densely populated areas like central Salem.
- Operations are typically restricted during peak hours and certain festival seasons to minimize risk.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Few recorded incidents involving hazardous materials, although historical data indicate occurrences of truck overturns due to road conditions and driver fatigue.
- Failures to adhere to speed limits in certain segments have resulted in collisions.

8. Environmental Considerations and Sensitive Areas

- Proximity to agricultural zones, necessitating careful management of any accidental spillage or contamination.
- Norms in place for minimizing noise and air pollution, particularly near eco-sensitive spots and water bodies.

9. Analysis of Communication Coverage

- Generally good mobile network coverage, though intermittent communication gaps may occur in more rural stretches.
- Potential dead zones around dense foliage and hilly parts near Salem outskirts.

10. Estimated Emergency Response Times for Different Route Segments

- Within Sangagiri or Sankari towns: Approximately 15-20 minutes.
- Rural road segments: Could extend to 30-45 minutes due to accessibility and response facility distances.
- In Salem urban areas: 10-15 minutes due to better strategic distribution of services.

11. Overall Summary of Risk Assessment

Risks Identified:

- Weather-related challenges predominantly during monsoon.
- Road quality affecting maneuverability, especially under loaded conditions.
- Notable congestion during peak times and specific local regulations regarding hazardous material transit.

Mitigation Strategies:

- Plan travel avoiding peak weather and traffic conditions.
- Regular vehicle checks and use of real-time navigation and communication tools to update route information.
- Ensure adherence to regulatory requirements and route-specific guidelines, especially regarding speed and hazardous material handling.

Overall, while the route is generally manageable, strategic preparation and real-time adaptability are key to ensuring the safety and efficiency of hazardous material transport on this path.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.67557, 78.15976	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.46048, 77.85036	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45634, 77.81673	15 KM/Hr
11	Turn	Medium	11.61748, 78.10604	30 KM/Hr
12	Turn	High	11.67074, 78.12668	15 KM/Hr
13	Turn	High	11.67081, 78.12661	15 KM/Hr
14	Turn	High	11.67582, 78.12786	15 KM/Hr
15	Blind Spot	Blind Spot	11.69586, 78.17683	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
21	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium
22	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
23	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium
24	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium
25	clinic	Arunagiri Ent Nursing Home	11.6762823, 78.1369593	30 km/h	Medium
26	clinic	Srimaya Child Health Centre	11.6750852, 78.1370646	30 km/h	Medium
27	hospital	Gl Hospital	11.6742597, 78.1379092	30 km/h	Medium
28	clinic	Ayush Multispeciality Homoeopathic Clinic	11.6770982, 78.1395947	30 km/h	Medium
29	hospital	Balaji Skin Hospital	11.676166, 78.1379488	30 km/h	Medium
30	hospital	Dr. Sasi Eye Care	11.6743698, 78.1379234	30 km/h	Medium
31	police	Alagapuram Police station	11.6769953, 78.1422615	30 km/h	Medium
32	hospital	AP medical center	11.6774074, 78.1425988	30 km/h	Medium
33	hospital	Dr. Soodamani Hospital	11.6757031, 78.14473	30 km/h	Medium
34	hospital	M G Diabetes Speciality and Research Centre	11.6766146, 78.1422986	30 km/h	Medium
35	hospital	Nathan Super Speciality Hospital	11.6743874, 78.1417222	30 km/h	Medium
36	hospital	Devi Hospital	11.6786391, 78.1415731	30 km/h	Medium
37	hospital	Fairlands Eye Cure	11.677959, 78.1415897	30 km/h	Medium
38	hospital	Kavin Eniyan Hospital	11.677643, 78.144129	30 km/h	Medium
39	hospital	Sri Vaari Hospital	11.6763726, 78.1467277	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
40	hospital	Krishna Speciality Hospital	11.676751, 78.145561	30 km/h	Medium
44	hospital	Gopi Memorial Hospital	11.673855, 78.1516078	30 km/h	Medium
45	clinic	Saraswathi Nursing Home	11.6746327, 78.1520981	30 km/h	Medium
46	hospital	J S Hospital, Dharmapuri	11.6739064, 78.1521557	30 km/h	Medium
47	hospital	Salem Gopi Hospitals	11.6745894, 78.1515482	30 km/h	Medium
48	hospital	Dr. P. Sundarrajan Neuro Hospital	11.6742791, 78.1499738	30 km/h	Medium
49	hospital	Mahadevas Ayurvedic Hospital	11.6752045, 78.1517396	30 km/h	Medium
50	clinic	Akshaya Fertility Centre	11.6739088, 78.1518657	30 km/h	Medium
52	hospital	Shanmuga Hospital and Salem Cancer Institute	11.6770178, 78.1531844	30 km/h	Medium
53	hospital	Tms Eye Hospital	11.6772791, 78.1536202	30 km/h	Medium
54	hospital	Sureka Hospital	11.676423, 78.1550511	30 km/h	Medium
56	hospital	Ramana Surgical Hospital	11.6731205, 78.15862	30 km/h	Medium
57	hospital	Salem Medical Centre	11.67607, 78.163014	30 km/h	Medium
58	hospital	KS Siddha Hospital	11.6752685, 78.1633063	30 km/h	Medium
59	police	Hastampatti B 7 Police Station	11.6778088, 78.1645798	30 km/h	Medium
60	hospital	Aishwaryam Nature Cure Hospital and Yoga Centre	11.6826829, 78.1758846	30 km/h	Medium
62	police	Kannankurichi B 8 Police Station	11.6967465, 78.1769759	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
16	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
41	school	Saradha College	11.6769525, 78.1481901	30 km/h	Medium
42	school	Saradha Balamandir School	11.6771182, 78.1472461	30 km/h	Medium
43	school	Sri Sarada Matriculation Higher Secondary School	11.6768969, 78.1488115	30 km/h	Medium
51	school	Holy flower matric hr.sec school	11.6748637, 78.1528099	30 km/h	Medium
55	school	CSI Hobart Edu Institute	11.673245, 78.1567934	30 km/h	Medium
61	school	Primery Health center kannankuruchi	11.6937205, 78.1755728	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67557, 78.15976



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61748, 78.10604



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67074, 78.12668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67081, 78.12661



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67582, 78.12786



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.69586, 78.17683